Soal Latihan UTS Mat Dis Variant: Answer Key

Answers for *UTS Practice Quiz Variant.pdf*. The methods are the same as in *UTS Quiz Solutions.pdf*, which has the full explanations and slide references.

Question 1

Method: The Logic of Compound Statements.pptx, slides 57–58, 64–70, 50.

a. Rows with S = 1 are 001, 011, 110 and 111.
S = P'Q'R + P'QR + PQR' + PQR

b. 001 = 1, 011 = 3, 110 = 6, 111 = 7.
S = Σm(1, 3, 6, 7)

c.

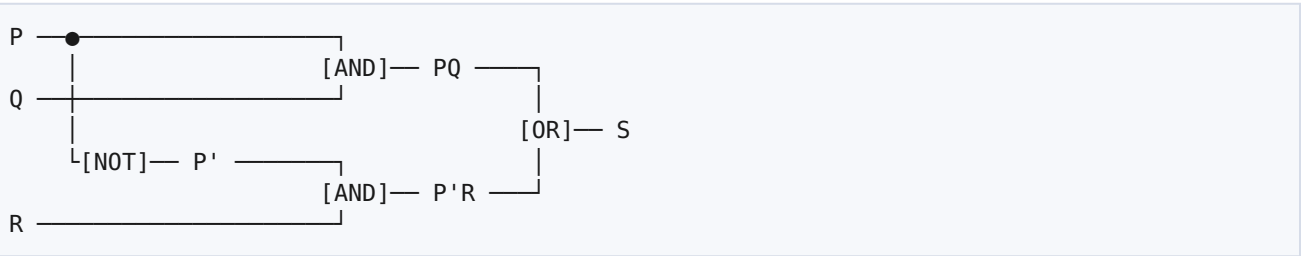
P \ QR	00	01	11	10
0	0 (m0)	1 (m1)	1 (m3)	0 (m2)
1	0 (m4)	0 (m5)	1 (m7)	1 (m6)

d.

- Group m1 + m3: P = 0 and R = 1 stay fixed, Q changes, giving **P'R**.
- Group m7 + m6: P = 1 and Q = 1 stay fixed, R changes, giving **PQ**.
- A group m3 + m7 (QR) is possible, but both of its cells are already covered, so it isn't needed.

S = P'R + PQ

e. Gates needed: 1 NOT, 2 AND, 1 OR.



Question 2

Method: Set Theory.pptx, slides 20–22.

a. 68 = 44 + 28 + 31 − 16 − 14 − 11 + x = 62 + x, so **x = 6**.
b. 120 − 68 = **52**
c. 16 − 6 = **10**
d. Venn regions, filled from the centre outward:

Region	Calculation	Value
M∩S∩L	(a)	6
M∩S only	16 − 6	10
M∩L only	14 − 6	8
S∩L only	11 − 6	5
M only	44 − 10 − 8 − 6	20

Region	Calculation	Value
S only	28 − 10 − 5 − 6	**7**
L only	31 − 8 − 5 − 6	12
Total		68 ✔

Only Science = 7

Question 3

Method: Properties of Functions and Fuzzy Set.pptx, slides 37, 39, 40.

x	μA	μB	A∪B = max	A∩B = min	A^c = 1 − μA
10	0.3	0.5	0.5	0.3	0.7
20	0.9	0.2	0.9	0.2	0.1
30	0	0.7	0.7	0	1
40	0.6	0.6	0.6	0.6	0.4
50	0.4	1	1	0.4	0.6

a. A∪B = 0.5/10 + 0.9/20 + 0.7/30 + 0.6/40 + 1/50
b. A∩B = 0.3/10 + 0.2/20 + 0/30 + 0.6/40 + 0.4/50
c. A^c = 0.7/10 + 0.1/20 + 1/30 + 0.4/40 + 0.6/50
d. μA(30) = 0, so 30 is left out. **supp(A) = {10, 20, 40, 50}**
e. Keep μA ≥ 0.4. The value 0.4 at x = 50 counts because the test is ≥. **$A_{0.4}$ = {20, 40, 50}**

Question 4

Method: Properties of Relation.pptx, slides 14–16, 30–38, 43–44.

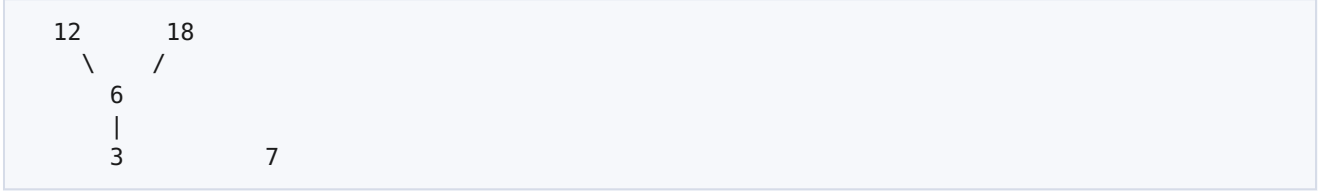
a. R = {(3,3), (6,6), (7,7), (12,12), (18,18), (3,6), (3,12), (3,18), (6,12), (6,18)}

b.

Property	Holds?	Justification
Reflexive	**Yes**	Every element divides itself.
Symmetric	**No**	(3,6) ∈ R but (6,3) ∉ R.
Antisymmetric	**Yes**	No two different elements divide each other.
Transitive	**Yes**	3|6 and 6|12 give 3|12 ∈ R. 3|6 and 6|18 give 3|18 ∈ R. Every other chain involves a loop.

c. Reflexive, antisymmetric and transitive, so **R is a partial order**.

d. Remove the loops and the arrows implied by transitivity, (3,12) and (3,18):



e.

- Minimal: **3, 7**

- Maximal: **7, 12, 18**
- Least: **None** (3 doesn't divide 7)
- Greatest: **None** (there are three maximal elements)